

How ChatGPT Assists Novices in Human Activity Recognition

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Abstract

It is evident that large language models are not only capable of generating text and images but also of facilitating discussions on the latest research in the field of human activity recognition due to their extensive knowledge. The objective of this paper is to observe how novices in machine learning in 2024 can rapidly develop their skills with the assistance of ChatGPT. As an initial introduction, we started with a dialogue-based consultation with ChatGPT and executed a classification task using the Iris dataset on Google Colaboratory. This initial task served to deepen the novices' basic understanding of machine learning. Subsequently, the authors, as part of Team Shonan-Blue, participated in the Sussex-Huawei Locomotion-Transportation (SHL) recognition challenge (hereafter SHL recognition challenge). In this challenge, the authors developed a simple human activity recognition system without expert advice and relying solely on interactions with ChatGPT. The system was constructed by integrating four classical methods: k-Nearest Neighbor algorithm (kNN), Neural Network (NN), Random Forest (RF), and Support Vector Machine (SVM). The models were selected for their differing approaches, which were anticipated to enhance prediction accuracy when combined. For feature extraction, we utilized only the primary four features (maximum, minimum, mean, standard deviation) derived from 5-second sensor data collected by accelerometers, gyroscopes, and magnetometers. The outputs of the four models were then processed ensemble to obtain the final prediction results. This yielded an F1 score of 0.936. Although the limited challenge period may not have permitted the optimal performance, the observed growth of novices who, with the assistance of ChatGPT and without the use of GPU power, managed to develop a functional system, is encouraging for future newcomers in the field of activity recognition.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Human-centered computing** → **Ubiquitous and mobile computing**.

Keywords

Human activity recognition; SHL dataset; ChatGPT

ACM Reference Format:

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1 Introduction

Human activity recognition using sensors is gaining increasing attention from various research communities. Team Shonan-Blue participated in the SHL recognition challenge with the goal of developing a human activity recognition algorithm using the SHL dataset.

Recently, advances in large language models (LLMs) have had a significant impact on activity recognition, as reported Kaneko [3] in HASCA 2023 Workshop and Sakai [4] in ABC Challenge 2024. For experts in activity recognition, LLMs are useful for exploring new features and investigating novel methods mentioned in recent research papers. On the other hand, the utility of LLMs for novice researchers and researchers from other fields has not been sufficiently addressed.

In this study, we closely observe the processes and outcomes when novices tackle activity recognition tasks with only the support of LLMs, without expert assistance. Specifically, we investigate how ChatGPT can support the initiation of activity recognition development skills. This research aims to provide valuable insights for researchers new to the field of activity recognition.

2 SHL dataset and task

The SHL dataset was collected primarily to investigate the recognition of users' activities from sensors in a mobile phone by means of machine learning methods and heuristics [2][7]. This dataset is a versatile annotated dataset of the modes of locomotion and transportation of users. It was recorded over a period of seven months in 2017 by three participants engaging in eight different modes of locomotion and transportation in a real-life setting in the United Kingdom. The dataset contains 750 hours of labelled locomotion and transportation data: Car (88 h), Bus (107 h), Train (115 h), Subway (89 h), Walk (127 h), Run (21 h), Bike (79 h), and Still (127 h). The dataset has multi-modal data from a body-worn camera and from four smartphones (HUAWEI Mate 9) carried simultaneously at typical body locations (bag, hips, torso, and hand).

The goal of the 6th SHL recognition challenge in 2024 is to recognize eight modes of locomotion and transportation (activities) in a user-independent manner based on motion sensor data in an opportunistic manner. In addition, the challenge data are prepared so that one or multiple sensor modalities are randomly missing from each time frame. The dataset used for this challenge comprised 59 days of train data and 28 days of test (challenge) data.

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The train data contains the raw sensors data (Acc, Gyr, and Mag) from one user (user 1) and four phone locations (bag, hips, torso, and hand). Each file contains 196,072 lines $\times$ 500 columns, corresponding to 196,072 frames each containing 500 samples (5 seconds at the sampling rate 100 Hz). The frames for the train data are consecutive in time.

The challenge data contains the raw sensors data from the other two users (mixing from both user 2 and user 3) and three phone locations (bag, hips, and torso). In each data frame, one or multiple sensor modalities are randomly missing by setting the sensor data to zero. The files contain 92,726 lines $\times$ 500 columns (5 seconds at the sampling rate 100 Hz).

3 Human activity recognition methods assisted by ChatGPT

This paper describes the process of a novice tackling human activity recognition. ChatGPT has an extensive knowledge of machine learning in general and human activity recognition in particular, and we sought to find an effective approach by using its help. Specifically, we describe how a novice who has completed initial training on a classification problem using the IRIS dataset can use that experience to take on the SHL Challenge.

Due to the short development schedule and limited computing resources, this study avoided using deep learning with GPUs. Although the latest ChatGPT (version 4 or later) has the ability to perform calculations internally, in this case, we chose to use the programming code recommended by ChatGPT and execute it in a different computing environment. This approach was aimed at improving the learning effectiveness of novices.

ChatGPT can understand the content of the challenge by specifying the URL of the SHL Challenge website. It can also analyze the content of a CSV file related to activity recognition. In order to obtain specific advice tailored to the user's level, initial input is required regarding experience with activity recognition and the computer environment to be used. A part of the chat log is shown in Figure 1.

Next, novices in human activity recognition were able to gain knowledge about building basic classification models through their interactions with ChatGPT. Below are some examples of questions used during the dialogue. A part of the chat log is shown in Figure 2, and 3.

What features should I use to recognize human activity?
What kind of model should I use?
What should I do if the number of samples between classes is different?

Next, the novice considered a policy of preparing a classification model for each sensor and obtaining prediction output by ensemble learning multiple models that exclude unusable sensors in order to improve robustness against missing data. This technique is called sensor fusion, and it combines different sensor data to improve prediction accuracy. The novice asked ChatGPT about this sensor fusion,

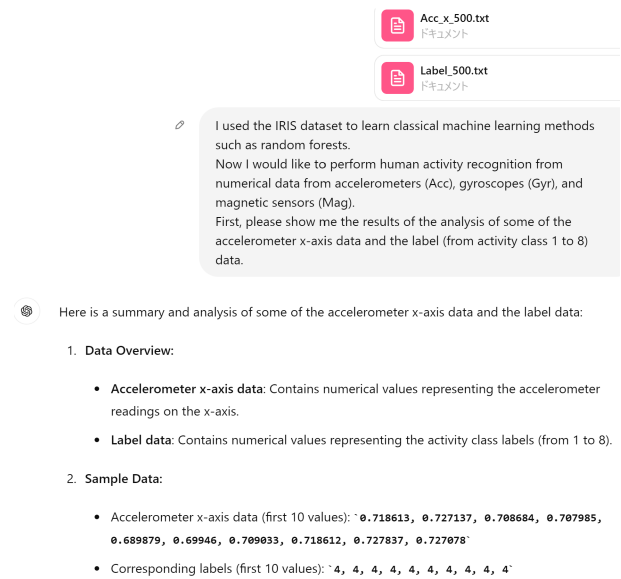


Figure 1: The beginning of the chat log with ChatGPT about data analysis of sensor data for human activity recognition.

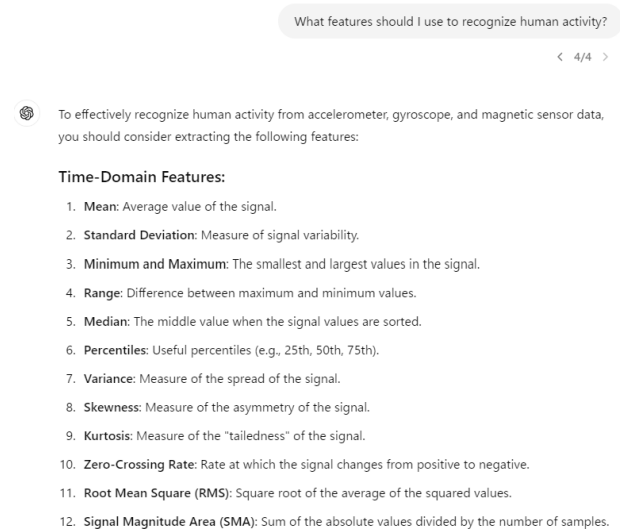


Figure 2: A part of the chat log with ChatGPT about selection of features.

I created prediction models for sensors Acc, Gyr, and Mag. However, one sensor (Acc) is broken. In this situation, how should I calculate the final prediction result?

We learned about the specific methods of ensemble learning. A part of the chat log is shown in Figure 4.

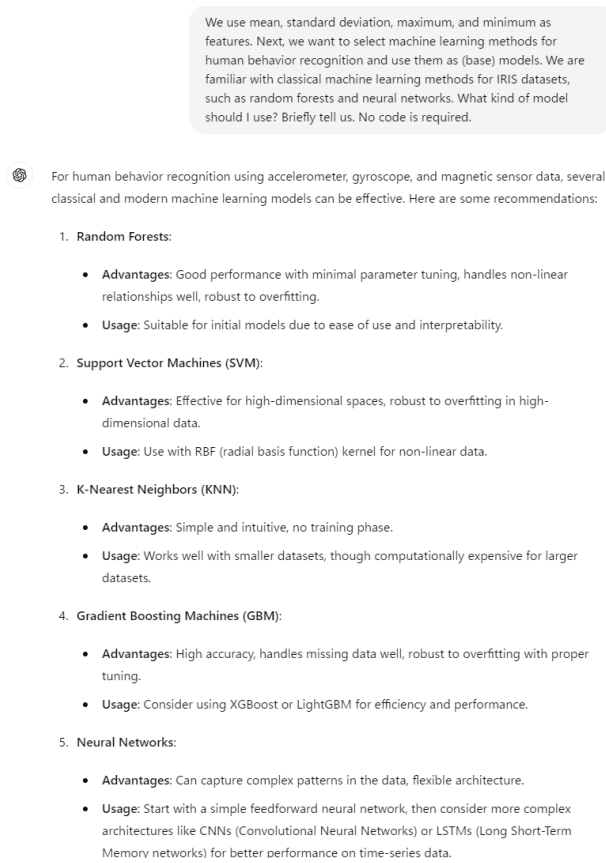


Figure 3: A part of the chat log with ChatGPT about selection of base models.

4 Experimental methods

4.1 Reconstructed dataset

As mentioned in the previous section, we obtained appropriate advice on activity recognition from ChatGPT and executed the Python code on Google Colab. The challenge data has sensors attached at one of the following locations: bag, hip, or torso. In order to create a classifier that is not dependent on the location of the sensor, we randomly sampled 500 examples for each of the eight activity classes from the labeled training data provided by the organizers, excluding the hand location (bag, hip, torso) sensors. Of these, 400 were used as the reconstructed training data, and the remaining 100 were used as the evaluation data. In addition, data in which activities transitioned within a 5-second period were excluded, and each frame of data was treated as having only a single activity label.

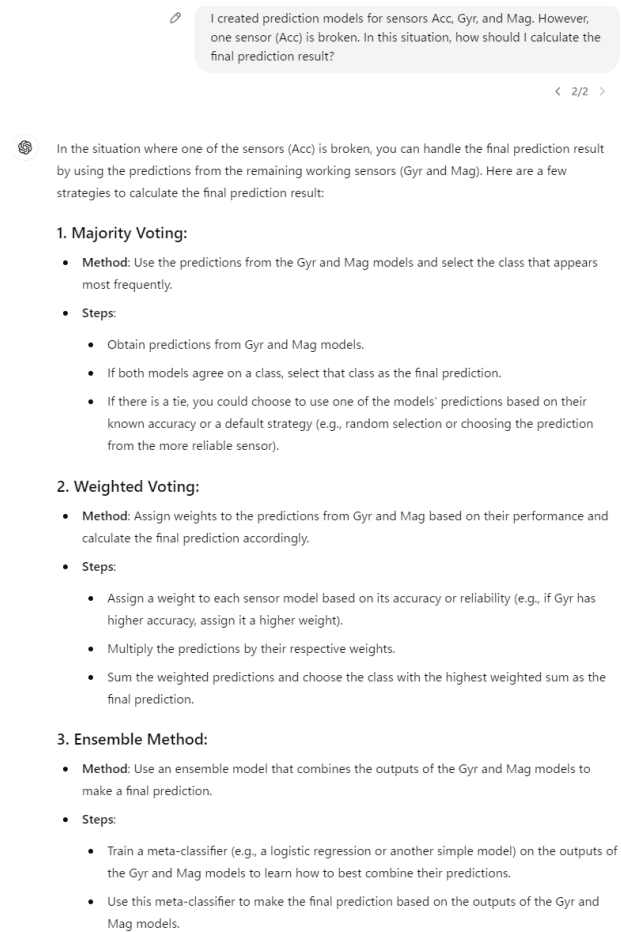


Figure 4: A part of the chat log with ChatGPT about integration of results of kNN, NN, RF and SVM.

Table 1: F1 scores of base models.

	kNN	NN	RF	SVM
Acc	0.831	0.844	0.872	0.700
Gyr	0.612	0.688	0.731	0.557
Mag	0.692	0.689	0.772	0.405

Table 2: Accuracy of base models.

	kNN	NN	RF	SVM
Acc	0.831	0.845	0.873	0.700
Gyr	0.628	0.689	0.730	0.568
Mag	0.701	0.690	0.774	0.413

Table 3: Confusion matrix of base models. From the left column, the models by kNN, NN, RF, and SVM are shown, respectively. From the top row, the models by Acc(1–4), Gyr(5–8), and Mag(9–12) sensors are shown, respectively.

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4.2 Basic classical machine learning models and sensor features

In order to cover a wide range of approaches, we adopted four basic classical machine learning models with different characteristics as the basic identification model: k-Nearest Neighbor algorithm (kNN), Neural Network (NN), Random Forest (RF), and Support Vector Machine (SVM). We extracted the average, variance, maximum value, and minimum value for each axis as feature values from the three types of sensors (accelerometer, gyroscope, and geomagnetic sensor) built into the smartphone for a period of five seconds. We randomly sampled 400 data points for each activity class and standardized them as a preprocessing step. We then trained the model and evaluated its performance using F1 score, accuracy, and confusion matrix.

4.3 Sensor fusion

In the challenge data, one of the three types of sensors could be missing, but two could not be missing at the same time. If a sensor is missing, all three axes of data for five seconds will be 0, so it is possible to identify missing data using a simple filter (for example, a filter that checks that all data is 0). Therefore, if there are any missing values, sensor fusion is performed using the base model corresponding to the two valid sensors, and if there are no missing values, sensor fusion is performed using the base model corresponding to the three sensors. Here, we performed stacking (the meta-model is Random Forest), which is one of the ensemble learning methods, to obtain the final prediction results.

5 Results

5.1 Performance of base models

We evaluated the performance of the four base models (kNN, NN, RF, and SVM) for the three types of sensors. The F1 score and accuracy are shown in Table 1 and 2. The confusion matrix is also shown in Table 3. Here, the rows indicate the true class, and the columns indicate the predicted class.

5.2 The effect of sensor fusion

The effectiveness of sensor fusion is shown using the same F1 score, accuracy (Table 4), and confusion matrix (Table 5) for cases where all sensors are normal, or where one of the accelerometer, gyroscope, or geomagnetic sensors is missing.

The evaluation data for cases where there are missing values uses the reconstructed data set described in subsection 4.1, and only uses the data from the two sensors that are not missing.

The F1 score for all data was 0.936, indicating high performance. Even when the gyro sensor was missing, the prediction accuracy did not decrease significantly, with only a 1.5 point decrease in F1 score. On the other hand, when the accelerometer was missing, the F1 score decreased by 8.7 points to a low of 0.849.

Table 4: F1 scores and accuracy of integrated models with missing data.

	Acc	Gyr	Mag	No missing
F1 score	0.849	0.921	0.915	0.936
Accuracy	0.850	0.921	0.915	0.936

Table 5: The confusion matrix is shown as the effect of sensor fusion when a sensor is missing. (1–3) show the case of missing Acc, Gyr, and Mag sensors, respectively. For comparison, (4) is the confusion matrix without missing sensors.

(1)

	1	2	3	4	5	6	7	8
1	81	1	0	1	1	13	3	0
2	1	94	0	3	0	2	0	0
3	0	2	97	0	0	0	1	0
4	0	2	0	96	0	1	0	1
5	1	0	0	0	88	8	2	1
6	6	0	0	1	3	84	4	2
7	3	0	0	0	8	5	62	22
8	0	0	0	0	2	3	17	78

(2)

	1	2	3	4	5	6	7	8
1	84	2	0	0	0	7	4	3
2	1	97	0	0	0	1	0	1
3	1	2	97	0	0	0	0	0
4	0	3	0	96	0	1	0	0
5	1	0	0	0	97	1	0	1
6	4	0	0	0	2	89	3	2
7	1	0	0	2	2	2	88	5
8	0	0	0	1	0	3	7	89

(3)

	1	2	3	4	5	6	7	8
1	81	2	0	0	0	2	9	6
2	1	97	0	0	0	1	0	1
3	1	2	97	0	0	0	0	0
4	0	2	0	96	1	0	0	1
5	0	0	0	0	98	1	0	1
6	1	0	0	0	1	88	3	7
7	3	1	0	0	2	2	88	4
8	3	0	0	0	0	5	5	87

(4)

	1	2	3	4	5	6	7	8
1	90	1	0	1	0	4	3	1
2	1	96	0	1	0	1	0	1
3	0	2	97	0	0	0	1	0
4	0	1	0	97	0	1	0	1
5	0	0	0	0	98	1	0	1
6	2	0	0	1	1	93	0	3
7	3	0	0	0	4	3	86	4
8	0	1	0	0	0	2	5	92

6 Discussion

Discussion for the base model: For all sensors, the RF model that was robust to feature variation showed the highest F1 score, and overall it showed good performance. Looking at each activity class, the sensor patterns for Class 3 (Run), which is a high-intensity activity, are clear and easy to classify for all models. On the other hand, there were many misclassifications for classes 6 (Bus) and 7 (Train), as their features often overlapped with those of other classes (5: Car, 6: Bus, 7: Train, 8: Subway). These results are similar to those of previous SHL challenges.

The results of sensor fusion by ensemble learning is a popular technique in the field of activity recognition [1][5][8]. We were able to derive reasonable performance gains as shown below; Discussion on Sensor Fusion: When there are no missing sensor data, using all the base models, we obtained higher F1 scores and accuracies as expected. When sensor data is missing, the prediction accuracy is lower than when there are no missing values, but there is a significant improvement compared to a single base model.

7 Conclusion

This paper examined the efficacy of utilizing ChatGPT as a tool for novices in machine learning in 2024. The results demonstrated that with the assistance of ChatGPT, they were able to significantly enhance their skills in a relatively short period of time. The novices from Team Shonan-Blue, who engaged in the SHL recognition challenge, achieved an F1 score of 0.936 in activity recognition through

ensemble learning with classical models such as kNN, NN, RF, and SVM. The recognition result for the test (challenge) dataset will be presented in the summary paper of the challenge[6].

ChatGPT not only provided advice on Python programming and machine learning platforms but also offered suggestions on suitable models, feature selection, handling imbalanced data, and post-processing multiple model outputs. Due to the limited period of approximately two weeks for development, the following issues remained: optimization of the window size, specification of the sensor positions, handling of individual differences, and robustness against missing data, which may have prevented optimal performance. Nevertheless, the progress made by novices who, with the assistance of ChatGPT and without the use of GPU power, managed to develop a functional system is a promising indication of the potential for future newcomers in the field of activity recognition.

Computer resources

We used a small notebook PC, the Let's Note (CF:CF-QV1FDMQR, CPU: 11th Gen Intel® Core™ i5-1135G7, 2.40 GHz, Memory: 16 GB) as the operating terminal. We accessed Google Colaboratory from this PC and performed all programming (Python 3) using Jupyter Notebook. We did not use a GPU. In addition, LLM used was ChatGPT Plus (Model GPT-4o).

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